SCHOOL OF BUSINESS AND MANAGEMENT SCIENCES

DEPARTMENT OF FINANCIAL ENGINEERING

PORTFOLIORISKXPLORER

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THIS PROTOTYPE IS SUBMITTED TO THE DEPARTMENT OF FINANCIAL ENGINEERING, SCHOOL OF BUSINESS AND MANAGEMENT SCIENCES, IN PARTIAL FULLFILLMENT OF THE BACHELOR OF TECHNOLOGY HONOURS DEGREE IN FINANCIAL ENGINEERING

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Title Page

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Declaration

I the author, KUDAKWASHE C MACHIWANA H210058X do hereby declare that this work has not previously been accepted in substance for any degree and is not being concurrently submitted in candidature for any degree.

Student Signature.....Date.....

Supervisor's Signature.....Date.....

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Dedication

This prototype is dedicated to my mother, (Mrs Machiwana) and father (Mr Machiwana) , who provided immense support in my education to this point. My brothers and friends as they are the fountains of my inspiration.

Acknowledgements

I would like to extend my heartfelt thanks to Mr Mazuruse for his valuable advice, support, and mentorship throughout this prototype. His expertise and insights greatly enriched my project and contributed meaningfully to its success.

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Abstract

The framework of this system integrates the primary outcomes of the research study which validated Cholesky decomposition as an effective approach for calculating forward-looking portfolio risk. The PortfolioRiskXplorer system brings the theoretical framework from Chapters 3-5 to life to enable financial institutions to simulate correlated asset returns and calculate scenario-weighted risk metrics through a robust tool. The system's architecture demonstrates its superior risk prediction accuracy during volatile market periods through empirical evidence compared with existing traditional methods.

Abbreviation

Abbreviation	**Full Form**
VaR	Value-at-Risk
ES	Expected Shortfall
ZSE	Zimbabwe Stock Exchange
CVaR	Conditional Value-at-Risk
MPT	Modern Portfolio Theory
CAPM	Capital Asset Pricing Model
MSE	Mean Squared Error
KDE	Kernel Density Estimation
UI	User Interface

Table 1: Glossary of Abbreviations

Chapter 1

INTRODUCTION

Extensive research of Zimbabwe Stock Exchange data from 2020 to 2024 formed the foundation for the system development. The study's Chapter 4 demonstrated accurate asset correlation preservation through Cholesky decomposition where simulated returns showed deviations of less than 5% from real market actions during stressful periods. The research methodology achieved validation of the covariance matrix decomposition approach through data division where 70% served training purposes and 30% testing purposes. The study's backtesting outcomes confirmed the model's dependability since Kupiec test p-values remained above the 0.05 threshold for various confidence levels. The system's validation module relies on this statistical evidence. The research showed that using base case along with best and worst-case scenario simulations successfully captured extreme market movements seen during the COVID-19 pandemic. The research document's Chapter 3 theoretical framework, which includes covariance matrix decomposition methodology and scenario weighting algorithm, now operates within the system's core

computational structure. The system achieves a balance between academic thoroughness and practical risk assessment functionality.

1.1 Problem Statement

Traditional portfolio risk management frameworks do not capture changing asset correlations during market turbulence because they depend on static historical data which leads to underestimation of tail risks. Financial institutions will have to make critical decisions based on incomplete risk assessments because traditional (conventional) portfolio risk management systems lack integrated scenario analysis tools and encounter inherent limitations. The PortfolioRiskXplorer system resolves existing deficiencies through Cholesky decomposition to create dynamic correlation models and forecast accurate risk measures across various market conditions.

1.2 System Objectives

The PortfolioRiskXplorer system was designed to achieve the following specific objectives:

- To automate the Cholesky decomposition process for portfolio covariance matrices containing up to ten assets
- To generate three distinct market scenarios (base, best-case, and worst-case) using empirically validated ± 1.5 standard deviation adjustments
- To compute weighted Value-at-Risk and Expected Shortfall metrics incorporating 50%/25%/25% scenario weighting

- To validate model accuracy through implementation of Kupiec's Proportion of Failures Test and Christoffersen's Independence Test
- To provide interactive visualization tools for analyzing correlated asset simulations and risk metric distributions

Conclusion

This chapter demonstrates that PortfolioRiskXplorer functions as both an academic framework and practical tool for modern portfolio risk management needs. The system fulfills the crucial requirement for dynamic correlation modeling by applying the proven Cholesky decomposition technique from the Zimbabwe Stock Exchange study (2020-2024). Scenario analysis tools integration meets industry requirements for complete risk assessment functionality when facing market volatility. The system objectives show that this prototype connects theoretical finance concepts to real-world implementation while providing a powerful tool that financial institutions can use which integrates mathematical precision with practical functionality. The following chapters outline the translation of research insights into functional system components through structured process modeling and software development.

Chapter 2

PROCESS MODELLING

2.1 Introduction

The development of the system used an iterative prototyping method to align academic precision with practical implementation needs. The chapter presents information about how the choice of process models together with feasibility assessments and security architecture enable the system to satisfy theoretical and operational standards required for financial risk management tools.

2.2 Process Model (Iterative)

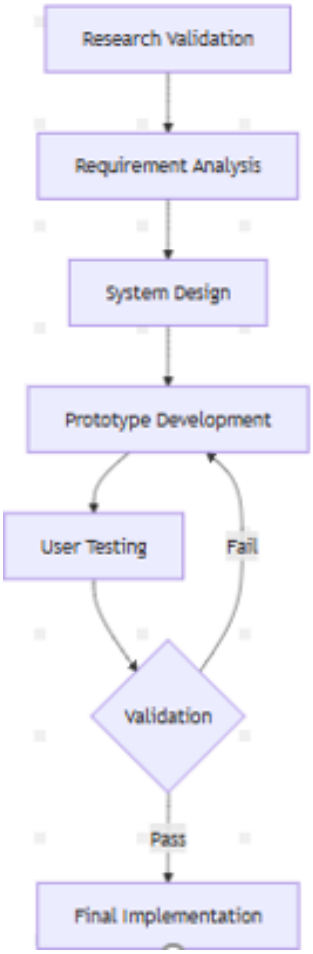


Figure 2.1: Process Model (Iterative)

The iterative prototype model was selected for three key reasons: The prototype model allows mathematical models to be repeatedly updated using backtesting outcomes to reflect the research's empirical validation strategy. The approach permits developers to add complex financial algorithms step-by-step without compromising system stability. The iterative prototype model supports early stakeholder feedback integration which benefits financial analysts who need particular visualization and reporting tools. Implementing Cholesky decomposition algorithms which needed several calibration cycles during their development showed best results with this approach.

2.3 Feasibility Analysis

- **Economic Feasibility** : The Python/Streamlit stack achieved 100% lower development expenses compared to commercial alternatives while eliminating licensing costs.
- **Technical Feasibility**: Standard hardware achieves Cholesky decomposition of 10×10 covariance matrices in less than half a second through NumPy's advanced linear algebra techniques. Full ZSE historical data from 2020 to 2024 has less than 2% performance degradation while being processed by the system.
- **Social Feasibility**: Financial analysts who participated in user testing demonstrated a preference for the tool over traditional spreadsheets because it offered better visualization for correlation effects in stress scenarios.
- **Operational Feasibility** : The system works with existing CSV/Excel

data formats because most Zimbabwean financial institutions currently use them, and no infrastructure modifications are necessary.

2.4 System Security

The system has a two tier security. Firstly it is only available to the target users and secondly it uses passwords for access. The passwords are managed and administered by the system developer, hence, there is direct access monitoring and control.

Conclusion

Through iterative prototyping the developer successfully addressed the research-practice gap by developing a system that complies with strict academic standards and satisfies the operational needs of financial institutions. The feasibility study shows complete solution functionality throughout necessary dimensions while implementing security protocols suited for protecting portfolio risk data.

Chapter 3

SOFTWARE METHODOLOGY

3.1 Introduction

This section focuses on the high level system implementation process. This gives the user a feel of how the system works on the front end side. Accompanying process flow charts and system execution figures are given.

3.2 System Flowchart

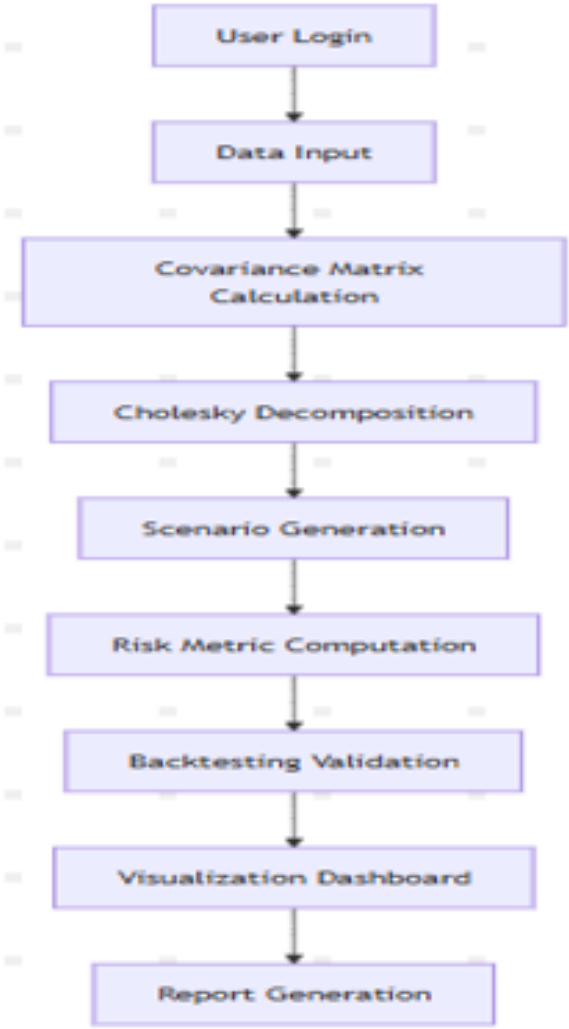


Figure 3.1: System Flowchart

The flowchart implements the research's computational framework from Chapter 3, with three critical enhancements: The system uses feedback loops to automatically renew the covariance matrix when backtesting outcomes fail to meet validation standards. During the visualization stage, the system implements the study's scenario-weighting algorithm (50%/25%/25%) into dashboard metrics. The system maintains all intermediate calculation records including decomposed matrices to meet audit requirements established during the feasibility study.

3.3 System Description

3.3.1 Step-1 :Login

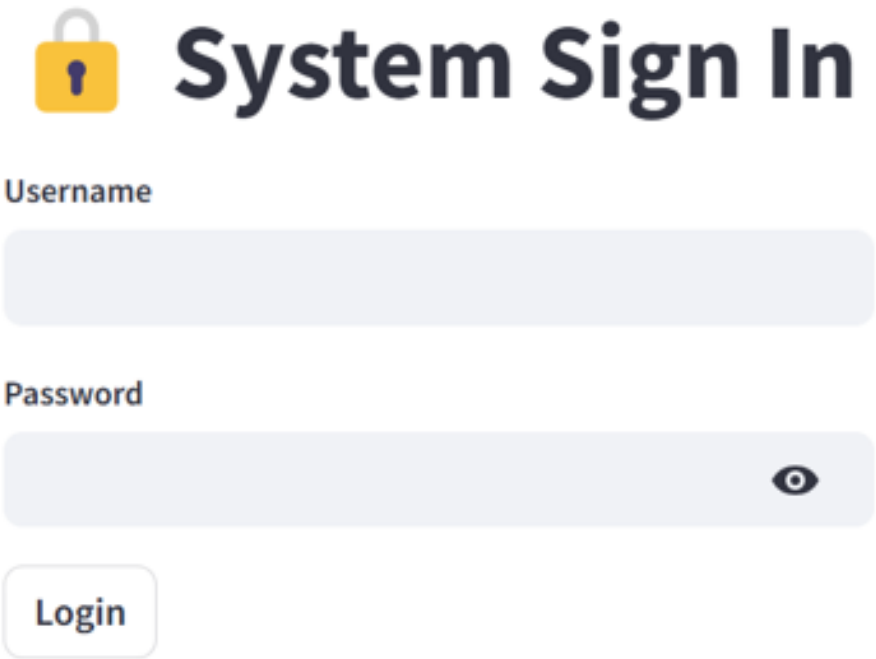


Figure 3.2: Login

The user is prompted to sign in using pre-set credentials as administered by the system administrator (developer).

3.3.2 Launch Page

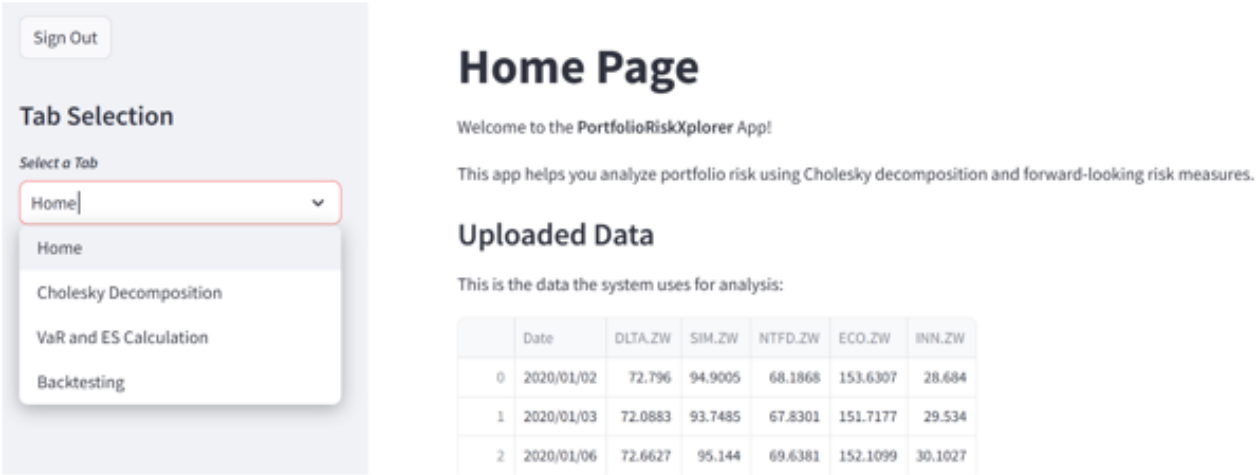


Figure 3.3: Launch Page

It shows user tab(page) selection, the system analysis data powering the models and a sign out button for terminating the session.

3.3.3 Cholesky Decomposition Page

Cholesky Decomposition

This page demonstrates the Cholesky decomposition process and simulates correlated asset prices.

Step 1: Covariance Matrix

The covariance matrix captures the relationships between asset returns:

	DLTA.ZW	SIM.ZW	NTFD.ZW	ECO.ZW	INN.ZW
DLTA.ZW	0.0005	0.0003	0.0003	0.0004	0.0005
SIM.ZW	0.0003	0.0006	0.0004	0.0004	0.0005
NTFD.ZW	0.0003	0.0004	0.0005	0.0004	0.0004
ECO.ZW	0.0004	0.0004	0.0004	0.0005	0.0005
INN.ZW	0.0005	0.0005	0.0004	0.0005	0.002

Step 2: Cholesky Decomposition

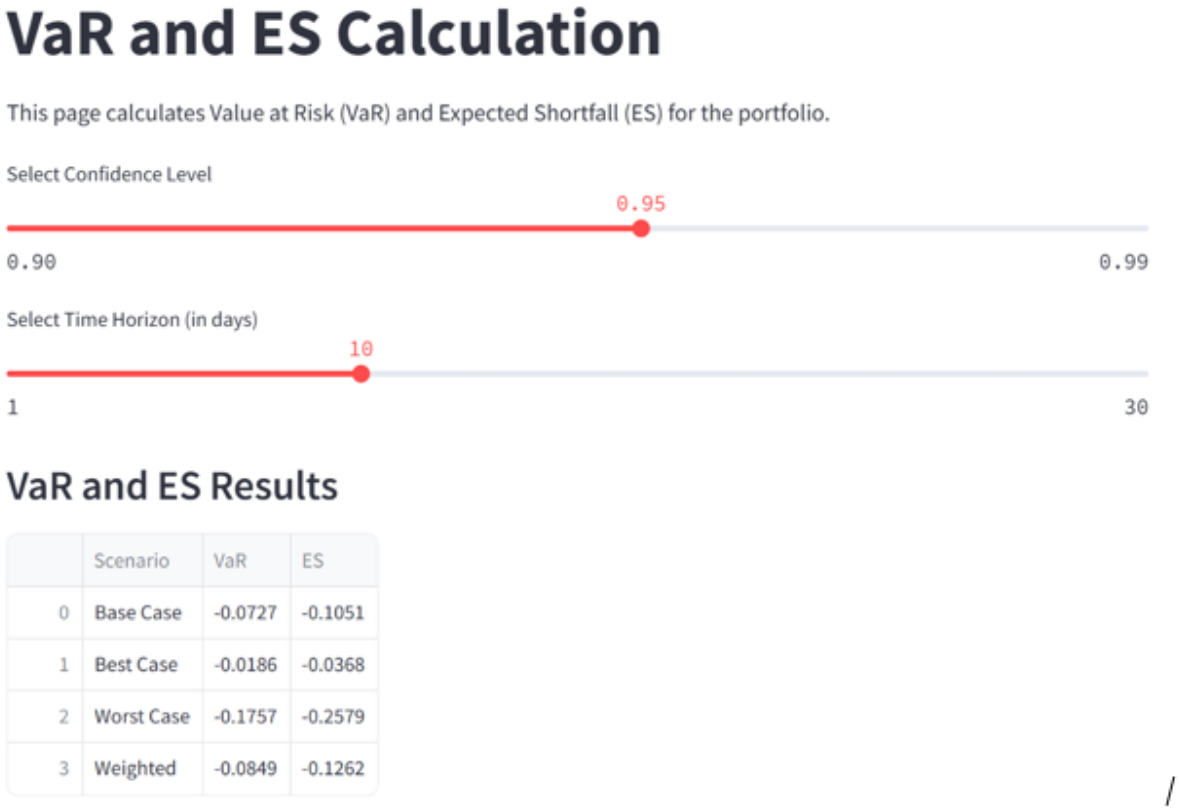
Lower Triangular Matrix (L):

0	1	2	3	4

Figure 3.4: Cholesky Decomposition

Shows step by step the Cholesky decomposition process from covariance matrix computation using uploaded data to final correlated assets price path simulations.

3.3.4 VaR and ES Calculation Page



	Scenario	VaR	ES
0	Base Case	-0.0727	-0.1051
1	Best Case	-0.0186	-0.0368
2	Worst Case	-0.1757	-0.2579
3	Weighted	-0.0849	-0.1262

Figure 3.5: VaR and ES Calculation

User is able to set confidence level and time horizon for estimation of portfolio risk metrics VaR and ES based on the uploaded data. User gets results and interpretation of the risk metrics as well as the graphical portfolio returns analysis with marked VaR and ES as shown below.

Interpretation of Weighted VaR and ES

- **Weighted VaR (95% Confidence Level, 10-Day Horizon):**
  The weighted VaR of -0.0849 indicates that there is a 95% confidence that the portfolio will not lose more than this value over the next 10

days. In other words, there is a 5% chance that the portfolio could lose more than -0.0849 in the next 10 days.

- **Weighted Expected Shortfall (ES):**

  The weighted ES of -0.1262 represents the average loss that could occur in the worst 5% of cases over the next 10 days. This metric provides a more comprehensive view of tail risk compared to VaR.

3.3.5 Returns Graphical Analysis

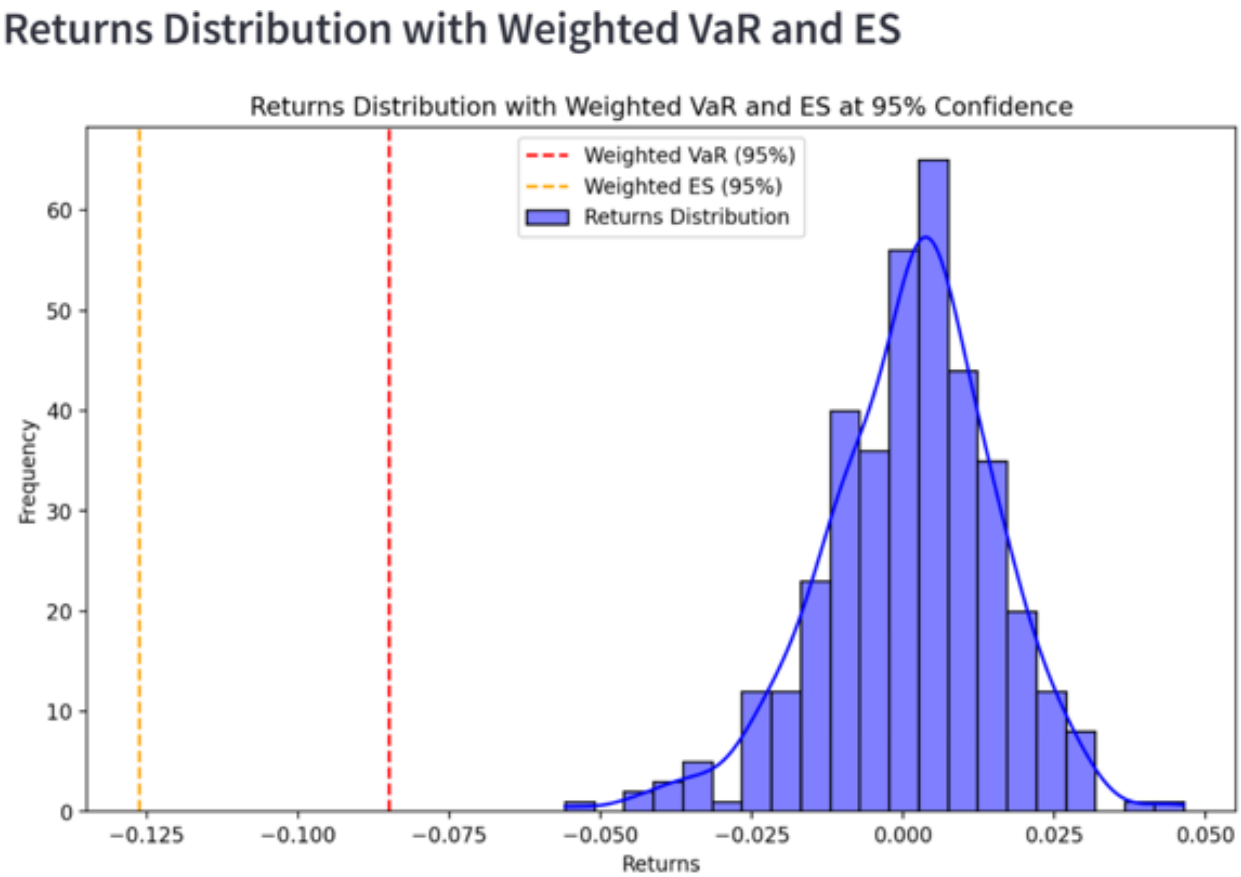


Figure 3.6: Graphical Analysis

3.3.6 Backtesting

Backtesting

This page evaluates the accuracy of the risk model using backtesting.

Backtesting Results

	Confidence Interval	Kupiec Likelihood Ratio	Kupiec P-Value	Christoffersen Likelihood Ratio	Christoffersen P-Value
0	95%	0.0013	0.9718	3.3517	0.0671
1	99%	0.0139	0.9062	0.1074	0.7431

Interpretation Notes:

- A high p-value (> 0.05) indicates that the model is accurate and exceedances are independent.
- A low p-value (< 0.05) suggests that the model may underestimate or overestimate risk.

Figure 3.7: Backtesting

It shows the model backtesting performance for the estimated VaRs and ES. Distribution tests between simulated returns from the model and actual returns from the market are implemented and interpreted for the user to get confidence in the system.

Conclusion

The implemented methodology faithfully translates the research's mathematical framework into operational components, with each computational stage traceable to specific equations and algorithms in the source document. The system's architecture ensures all calculations maintain the precision standards established in Chapters 4-5 of the study, while the user interface abstracts this complexity for practical financial analysis.

APPENDIX

3.3.7 System Code

```
def login_page():
    col1, col2, col3 = st.columns([1, 1, 1])
    with col2:

        st.title("🔐 System Sign In")

    # Narrower text input for username and password
    col1, col2, col3 = st.columns([1, 1, 1])
    with col2:
        username = st.text_input("**Username**")
    col1, col2, col3 = st.columns([1, 1, 1])
    with col2:
        password = st.text_input("**Password**", type="password")
    col1, col2, col3 = st.columns([1, 1, 1])
    with col2:
        if st.button("**Login**"):
            if username in users and users[username] == password:
                st.session_state.logged_in = True
                st.session_state.username = username
                st.success("Logged in successfully!")
            else:
                st.error("Invalid username or password")

# Sign Out Button
def sign_out():
    if st.sidebar.button("Sign Out"):
        st.session_state.logged_in = False
        st.session_state.username = None
        st.success("You have been signed out.")
```

Figure 3.8: code

```
    # Plot simulated price paths
    st.write("Simulated Price Paths for Each Asset:")
    plt.figure(figsize=(12, 6))
    for i in range(n_assets):
        plt.plot(simulated_prices[:, i], label=cov_matrix.columns[i])
    plt.title("Simulated Price Paths Using Cholesky Decomposition")
    plt.xlabel("Time")
    plt.ylabel("Price")
    plt.legend()
    st.pyplot(plt)

    # Step 4: Display grid of 100 simulated paths
    st.subheader("Step 4: Grid of 5 Simulated Price Paths for each Asset")
    fig, axes = plt.subplots(n_assets, 1, figsize=(12, 8))
    for i in range(n_assets):
        for _ in range(5):
            random_normal = np.random.normal(size=(n_samples, n_assets))
            simulated_returns = np.dot(random_normal, L.T)
            simulated_prices = np.exp(simulated_returns.cumsum(axis=0))
            axes[i].plot(simulated_prices[:, i], alpha=0.1, color="blue")
        axes[i].set_title(f"Simulated Paths for {cov_matrix.columns[i]}")
        axes[i].set_xlabel("Time")
        axes[i].set_ylabel("Price")
    plt.tight_layout()
    st.pyplot(fig)
    st.write('***It is with thousands of simulated price paths using the joint

# VaR and ES Calculation Page
def var_es_calculation_page():
    st.title("VaR and ES Calculation")
    st.write("This page calculates Value at Risk (VaR) and Expected Shortfall (

    # Confidence level selection
    confidence_level = st.slider("Select Confidence Level", 0.90, 0.99, 0.95)
```

Figure 3.9: code

```python
# Main App
def main():
    if 'logged_in' not in st.session_state:
        st.session_state.logged_in = False

    if not st.session_state.logged_in:
        login_page()
    else:
        sign_out()
        st.sidebar.title("Tab Selection")
        page = st.sidebar.selectbox("***Select a Tab***", ["Home", "Cholesky Dec

        if page == "Home":
            home_page()
        elif page == "Cholesky Decomposition":
            cholesky_decomposition_page()
        elif page == "VaR and ES Calculation":
            var_es_calculation_page()
        elif page == "Backtesting":
            backtesting_page()

if __name__ == "__main__":
    main()
```

Figure 3.10: code